

Remimazolam versus midazolam for procedural sedation in children undergoing MRI: a randomised, double-blind, non-inferiority trial

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Background

Procedural sedation for paediatric magnetic resonance imaging requires rapid onset and predictable recovery. Remimazolam, an ultra-short-acting benzodiazepine, may offer advantages over midazolam.

Methods

We randomised 120 children aged 3-12 years (ASA I-II) to remimazolam (n=61) or midazolam (n=59). The primary outcome was sedation success without rescue. Non-inferiority margin was 10 percentage points.

Results

Sedation success was 93.4% (57/61) with remimazolam versus 79.7% (47/59) with midazolam (risk ratio 1.24, 95% CI 1.05-1.47; $p=0.012$). Median recovery time was 12.4 min (IQR 9.8-15.1) versus 24.7 min (IQR 19.2-31.5). Hypotension occurred in 3 (4.9%) versus 9 (15.3%) children. No serious adverse events.

Conclusions

Remimazolam was non-inferior and superior to midazolam for sedation success, with faster recovery and fewer haemodynamic events. Registration: NCT04412345.